Samuel Dawit Assefa

B.S. in Computer Science, KAIST (Senior, 7th Semester)

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Research Interests

Alignment and interpretability of parameter-efficient language models; augmentation and distillation; calibration and uncertainty; AI safety.

Education

Korea Advanced Institute of Science and Technology (KAIST) **Daejeon, South Korea**
B.S. in Computer Science 2023–2027

- Relevant coursework: Statistical Machine Learning; Artificial Intelligence and Machine Learning; Introduction to Artificial Intelligence; Introduction to Deep Learning; Probabilistic Machine Learning.

Publications & Workshop Papers

Hidden Commitment: When Language Models Silently Pick a Side and How Steering Can Surface It **2026**
*First author — **Accepted** at the AI4GOOD Workshop, ICML 2026 (Seoul)*

- Identified a **hidden source-commitment failure**: models can internally choose one conflicting document while answering as if the evidence were settled.
- Learned a residual **source-choice direction** with **0.994 held-out AUROC** in Qwen 0.8B and **0.980 AUROC** transfer to Gemma 3 1B on 907 audited rows.
- Localized sparse MLP writers feeding the same coordinate; CETT neurons predicted held-out chosen source with **0.900 AUROC / 0.808 F1**.
- Demonstrated causal control: steering shifted answered source bidirectionally; disclosure steering raised conflict surfacing from **0% to 17.7%**, and a two-direction intervention to **45.1%**.

Feature-Based Prediction of DNA Fragmentation in Live Mouse Sperm Using Label-Free Holotomography **2026**
First-author conference abstract/poster — Biophysics Division, Korean Physical Society

- Presented a label-free AI framework for **single-cell prediction of post-imaging TUNEL labels** from live 3D refractive-index tomograms.

Research Experience

PuzzleAI **Daejeon, South Korea**
AI Engineer Intern; Mentor: Harin Jun, CTO Dec 2025–Feb 2026

- Built **automated LLM evaluation benchmarks** for clinical-domain outputs (LLM-as-judge + rule-based checks + regression tests), with **hallucination metrics** reaching $>$**90% agreement** with gold labels at $<$**2% variance**.
- Re-architected a clinical coding/retrieval pipeline (recursive tree-walk RAG $\rightarrow$ **direct candidate generation + local hierarchy refinement**), raising exact F1 by **17.5%** and refinement precision by **55.2%** while cutting predicted labels by **37.2%**.
- Cut benchmark runtime $>$**50**$\times$ (minutes $\rightarrow$ seconds per case), enabling controlled model/pipeline ceiling comparisons.

BioMedical Optics Lab, KAIST **Daejeon, South Korea**
Researcher in Computer Vision and Data Science; Advisor: YongKeun (Paul) Park Jul 2025–Present

- Built a label-free CV/data-science pipeline for **live single-cell sperm DNA-fragmentation prediction** from 3D holotomography.
- Developed segmentation and label-curation (heuristic sperm-head isolation + **transfer-finetuned models**), reducing **1,948 to 1,039 higher-confidence cells** and extracting **41 refractive-index features**.
- Trained interpretable ElasticNet models reaching **ROC-AUC 0.739 / PR-AUC 0.648**; strongest signals came from RI texture, skewness, and radial organization.

Personal Projects

Can Sparse Neuron Subspaces Warn You Before Safety Breaks? **2026**
Independent AI safety research

- Asked whether a small set of neurons can act as an **early-warning signal for safety degradation**: localized a sparse (∼**156-neuron**) subspace separating resistance vs. compliance under misleading context.
- During benign fine-tuning, this subspace drifted *before* any measurable behavioral change — a few unusually **training-sensitive neurons** that appear to track safety drift early. Exploratory, but a promising direction for fine-tuning monitors.

Representation of Document Choice Selection in Conflicting-Context Language Models **2026**
Self-directed mechanistic-interpretability project Blog write-up

- Explored whether document choice under contradiction is represented as a **low-dimensional residual-stream state**.
- Used **document-order swaps** and answer-span timing analyses to separate selected-source representations from truth, first-document, and lexical-answer confounds.
- Wrote up the investigation as a public blog series; this representation-analysis thread later became the source-choice component of *Hidden Commitment*.